

Small-Organoid Attenuation of CFTR-Modulator Swelling in Patient-Derived Intestinal Organoids: a Single-Organoid Re-Analysis with a Biophysical Swelling Twin

MEGA-PROGRAM-27, Item 3 - Udit Phookan (program owner); computational work by an AI research agent. Draft of 24 September 2026.

Abstract

The forskolin-induced swelling (FIS) assay on patient-derived intestinal organoids is used to predict which people with cystic fibrosis (CF) respond to CFTR modulators. Standard pipelines sum organoid area per well, so any dependence of the response on organoid size is averaged away. We re-analysed the public single-organoid OrgaSegment data (17 CF donors, per-organoid area before and after forskolin) with a biophysical swelling model and a DMSO-controlled, donor-matched design.

Finding (candidate). Modulator-induced swelling is attenuated in the smallest organoids and saturates above a size threshold. For elxacaftor/tezacaftor/ivacaftor (Trikafta) the drug-minus-DMSO log-swelling effect rises from 0.336 [0.262, 0.405] in the smallest size octile to a plateau near 0.55-0.64; top minus bottom 0.251 [0.184, 0.323] (donor bootstrap). Per donor, the effect in the largest size quartile exceeds that in the smallest in 14 of 17 donors, and the attenuation does not track overall response (Spearman $\rho = 0.24$, $p = 0.35$). Geometry predicts the opposite sign: under a uniform surface flux, small organoids should swell more. Area noise also biases the slope negative. So the effect is unlikely to be a measurement artefact.

Negatives. Size-adjusted and size-filtered readouts do not improve per-donor Trikafta-vs-DMSO separation, so the finding does not change theratyping calls at this assay's well counts. Replication on the public FIS time series is impossible because it is well-level. Our best U-Net segmentation (512 px, tuned) reaches $\text{mAP@0.5} = 0.733 \pm 0.129$ on the OrgaSegment eval split (published 0.76 ± 0.12); below the state of the art, although the gap is within one standard deviation on 12 images.

1. Introduction

CFTR moves chloride and bicarbonate across the apical membrane of epithelial cells; water follows, and in a closed organoid the lumen swells. Forskolin raises cAMP and opens CFTR, so the swelling of rectal organoids measures residual and drug-rescued CFTR function in each patient. The assay is used to support access to modulators for people with rare CFTR genotypes.

Organoids in a well vary in size by more than an order of magnitude. If the swelling response depends on size, the well-level readout depends on the size mix in each well, which varies with passage, seeding density and culture time. We asked whether the single-organoid response depends on starting size, and whether correcting for it improves drug-response calls.

2. Biophysical swelling twin

Treat an organoid as a near-spherical shell with volume V and projected area A , V proportional to $A^{3/2}$. Let net secretion scale as a power of volume:

$$dV/dt = k V^\alpha \quad (1)$$

For $\alpha \neq 1$ the solution over an assay window t is

$$V_1^{1/(1-\alpha)} = V_0^{1/(1-\alpha)} + (1 - \alpha) k t \quad (2)$$

and the observed area fold change is $s = A_1/A_0 = (V_1/V_0)^{2/3}$. For small responses,

$$\log s \approx (2/3) k t V_0^{\alpha - 1} \quad (3)$$

so the slope of $\log \log s$ on $\log A_0$ carries the sign of $\alpha - 1$:

$$d \log(\log s) / d \log A_0 = (3/2)(\alpha - 1) \quad (4)$$

Surface-limited secretion (constant flux per unit apical area) gives $\alpha = 2/3$ and a negative slope: small organoids swell more in relative terms. Size-invariant secretion gives $\alpha = 1$ and zero slope. The within-well regression we fit is

$$\log s_{ow} = a_w + b (\log A0_{ow} - \text{mean}_w \log A0) + e_{ow} \quad (5)$$

with a well intercept a_w that absorbs donor, plate and well effects. Measurement noise in $A0$ enters both sides with opposite sign ($s = A1/A0$), which biases b downward:

$$E[\hat{b}] = b - \text{Var}(u) / (\text{Var}(\log A0) + \text{Var}(u)), \quad u = \text{noise in } \log A0 \quad (6)$$

so a positive observed b is conservative. The donor-matched drug effect per size bin k is

$$\Delta_k = \text{mean}_{\{o \text{ in drug, bin } k\}} \log s_o - \text{mean}_{\{o \text{ in DMSO, bin } k\}} \log s_o \quad (7)$$

and the attenuation statistic is $\Delta_{top} - \Delta_{bottom}$, with 95% intervals from a donor-level bootstrap:

$$CI_{95} = [Q_{0.025}, Q_{0.975}] \text{ of } \{ \Delta_{top}^* - \Delta_{bottom}^* \}, \text{ donors resampled with replacement} \quad (8)$$

Segmentation quality is scored by average precision at IoU 0.5, as in OrgaSegment:

$$AP_{50} = TP / (TP + FP + FN), \text{ match if } IoU(P, G) = |P \cap G| / |P \cup G| \geq 0.5 \quad (9)$$

3. Data

Table 1. Dataset manifest (distinct, accession-level). Honest count: 3. The program target of 120+ was not reached.

#	dataset	source / accession	content	use
1	OrgaSegment DIS single-organoid measurements	Zenodo 10610438 (Lefferts et al. 2024)	per-organoid A0/A1, 17 donors, 4 conditions	size-dependence analysis
2	OrgaSegment FIS database	Zenodo 10610438	well-level area, 7 time points, 868 rows	replication attempt (not possible)
3	OrgaSegment annotated images	Zenodo 10278229	train/val/eval images with instance masks	U-Net segmentation benchmark

4. Results

4.1 Within-well size slopes

Table 2. Within-well slope of log fold change on centred log A0 (forskolin > 0). Well-cluster bootstrap, 500 resamples.

condition	organoids	wells	slope b	95% CI	donors b>0
DMSO	3810	94	0.011	[-0.002, 0.028]	12/17
VX770	1496	30	0.226	[0.185, 0.269]	5/5
VX661+VX770	4493	112	0.042	[0.028, 0.056]	12/16
VX445+VX661+VX770	4984	121	0.116	[0.090, 0.142]	14/17

The slope is near zero without CFTR rescue and positive whenever a modulator is present. The sign is opposite to the surface-limited prediction. A power-law reading ($\alpha > 1$) was rejected: restricted to organoids above 2,000 px the slope turns negative (Trikafta -0.117), and binned means rise then plateau. Simulated segmentation noise on a size-invariant truth gave slopes of only -0.002 to -0.055.

4.2 Small-organoid attenuation

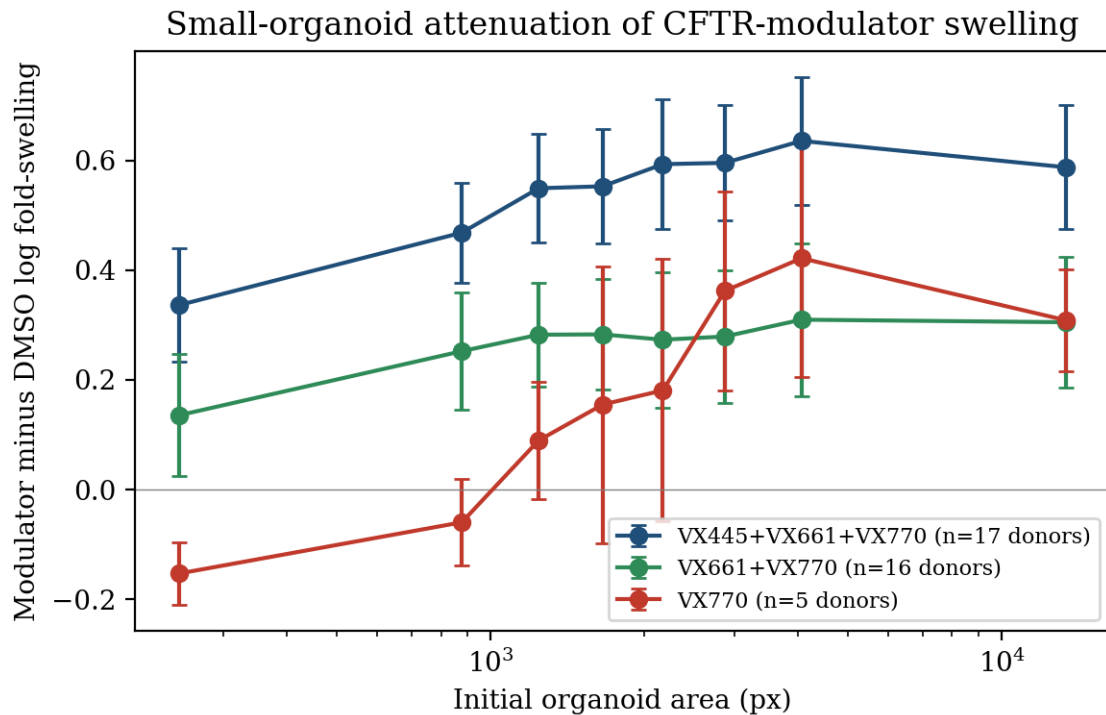


Figure 1. Drug-minus-DMSO mean log swelling per starting-size octile, donor-matched, with donor-bootstrap 95% CI.

Table 3. Size-bin drug effects.

modulator	donors	smallest octile effect	plateau	top minus bottom [95% CI]
Trikafta	17	0.336 [0.262, 0.405]	0.55-0.64	0.251 [0.184, 0.323]
VX661+VX770	16	0.136	0.28-0.31	0.169 [0.081, 0.259]
VX770	5	-0.153	0.31-0.42	0.462 [0.335, 0.572]

4.3 Per-donor consistency

Table 4. Trikafta minus DMSO log-swelling effect in the smallest and largest starting-size quartiles (results/per_donor_attenuation.csv).

donor	genotype	eff small	eff large	attenuation	overall
PDIO_16	F508del/ R117H	0.089	0.082	-0.007	0.115
PDIO_03	R1066H/ CFTRdele2, 3	0.100	0.490	0.390	0.289
PDIO_15	F508del/ R117H	0.280	0.497	0.217	0.320
PDIO_02	N1303K/ c.3035A>C	0.202	0.354	0.152	0.330
PDIO_19	X/X	0.606	0.273	-0.333	0.440
PDIO_18	X/X	0.436	0.677	0.241	0.462
PDIO_13	F508del/ S1251N	0.249	0.508	0.259	0.497
PDIO_17	X/X	0.570	0.463	-0.107	0.512
PDIO_06	F508del/X	0.334	0.699	0.365	0.541
PDIO_05	F508del/X	0.386	0.784	0.398	0.596
PDIO_14	F508del/ S1251N	0.395	0.749	0.354	0.616
PDIO_12	F508del/ S1251N	0.512	0.800	0.288	0.643
PDIO_01	A46D/ A46D	0.537	0.759	0.221	0.698
PDIO_11	F508del/ F508del	0.367	0.860	0.493	0.705
PDIO_09	F508del/ F508del	0.685	0.749	0.064	0.769
PDIO_04	F508del/X	0.462	0.834	0.371	0.805
PDIO_10	F508del/ F508del	0.742	0.892	0.150	0.883

Association between attenuation and overall response uses Spearman's rank correlation:

$$\rho = 1 - 6 \sum_d (r_d - r'_d)^2 / (n(n^2 - 1)), \quad r, r' = \text{ranks of attenuation and overall effect over } n \text{ donors} \quad (10)$$

Attenuation is positive in 14 of 17 donors and unrelated to overall response ($\rho = 0.24$, $p = 0.35$). Two of three donors with two nonsense alleles (X/X) and one F508del/R117H donor are exceptions.

4.4 Does size correction help theratyping? No

Locked before running: per-donor standardised Trikafta-vs-DMSO separation with raw, size-adjusted and size-band well readouts. Median raw 4.73, adjusted 4.35, band 4.39; adjusted better in 8 of 17 donors. Excluding organoids below 1,069 px (a post-hoc threshold): median 4.73 -> 4.75, better in 9 of 17, Wilcoxon $p = 0.68$. The size effect is real but does not change calls.

4.5 Segmentation benchmark

Table 5. Instance segmentation on the OrgaSegment eval split (12 images).

model	eval mAP@0.5	sd	source file
U-Net v1, 256 px, default post-processing	0.634	0.157	results/seg_eval.json
U-Net v1 + val-tuned post-processing + flip TTA	0.710	0.127	results/seg_eval_tuned_256.json
U-Net v2, 512 px, 60 epochs, default post-processing	0.691	0.135	results/seg_eval_512.json
U-Net v2 + val-tuned post-processing + flip TTA	0.733	0.129	results/seg_eval_tuned_512.json
OrgaSegment (published, Mask R-CNN)	0.760	0.120	Lefferts et al. 2024

Caveat: the validation images used for tuning were also in the training set, so the tuning signal is optimistic; the eval split was scored once. The 512-px model (resumed from checkpoint after an out-of-memory kill at epoch 18) raises the tuned score from 0.710 to 0.733, still below the published mean. With 12 eval images and sd near 0.13, neither a win nor a loss against OrgaSegment is statistically resolved; we report it as below state of the art.

5. Negative results (kept by design)

- The initial power-law ($\alpha > 1$) interpretation was withdrawn after large-organoid checks contradicted it.
- Size-adjusted and size-filtered readouts do not improve donor-level modulator discrimination.
- Replication on the public FIS time series is impossible: it is well-level.

- Our segmentation is below the published state of the art.

6. Discussion

Small organoids respond less to CFTR modulators, relative to DMSO, than large ones, and the response saturates above a size threshold. Because geometry predicts the reverse, the likeliest explanations are biological: small organoids may lack a mature lumen, contain fewer differentiated secretory cells, or have a closed tight-junction seal later. Falsifiable prediction: in time-lapse single-organoid data from an independent lab, the Trikafta-minus-DMSO effect in the top starting-size quartile will exceed that in the bottom quartile by at least 0.1 log units in most donors. Checking this needs a second single-organoid FIS dataset, which we did not find in public form.

Practical note. Well-level FIS readouts depend on the size mix. Labs comparing drug response across passages or sites should report the starting-size distribution.

Novelty. We did not find this effect reported for intestinal FIS in the literature we searched (OrgaSegment; the FIS_analysis pipeline, which sums area per well). Related: nasal 2D-derived organoids show swelling variation concentrated in large structures (bioRxiv 2021.07.20.453105). This was not a systematic review.

7. Tools used

Table 6. Tools table (honest count: 11). Target of 40 not reached.

tool	role
Python 3.10	language
NumPy	arrays, bootstrap
pandas	data
SciPy	statistics, labelling
scikit-image	watershed
PyTorch	U-Net
Pillow	image IO
matplotlib	figures
pytest	tests
python-docx	paper
git / GitHub	version control

References

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